

This is part of a manuscript describing a model to test the bacterial burden in a system based on the changing the antibiotic concentration. This figure shows a compilation of the bacterial burden based on a given EC50. we are trying to determine if this is the best way to present the data. This manuscript will probably be submitted to a math journal but I want to bridge the gap between math and biology people. is there a more effective way to present the data? The most important part of the figure is the "switch" (0.988) which is the EC50 where the switch occurs. Below this value, there is less bacterial burden because the drug is more effective, etc

Architected concise yet substantive explanation of phase diagram visualization

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Here's what that would look like, using dose as the second axis. Note that the boundary shape here is invented — I don't have your model — but the layout is what matters:

The single most valuable annotation is that dashed horizontal line. It shows that your entire current Figure 1 is one horizontal slice through this plane, and 0.988 is just where that slice crosses the boundary. A reader who understands that will immediately grasp what the threshold does and doesn't mean — it isn't a property of the drug, it's a property of the drug *at that dose*. For a biology audience that reframing is the whole payoff, and for a math audience it turns a number into a curve. Some things worth thinking about as you build it:

The shape of the curve is your actual result. I drew it exiting through the top of the plot, which would mean that beyond $EC_{50} \approx 1.6$ no dose in the tested range suppresses. Whether your boundary does that, or bends over and saturates, or has a vertical asymptote, is exactly the finding a math reviewer will want. A vertical asymptote in particular would be a strong claim: some EC_{50} beyond which escape is unavoidable at any dose.

Don't compute this on a grid. A 6×4 grid is 24 simulations and gives you a staircase. Instead, for each dose on the y-axis, bisect on EC_{50} to find the critical value — maybe 12 bisections per dose, 15 doses, and you get a smooth curve with a stated tolerance for roughly the same compute as a coarse grid. Then show the grid points only sparsely, as evidence, the way I've drawn them. Report the tolerance in the caption.

Consider a heatmap underneath. If terminal burden varies continuously within each region (not just two levels), fill the plane with a sequential colormap of $\log_{10}$ burden and overlay the boundary as a black contour. That shows both the sharp transition and any gradient structure. If burden really is bimodal, skip the heatmap — two flat shaded regions are more honest.

Which second parameter to pick. Dose is the most clinically legible and the easiest for a biology reader to act on. Dosing interval is a close second and lets you make a point about

AUC-versus-time-above-threshold that pharmacologists care about. Reservoir exchange rate is the most interesting mathematically, since it's the coupling parameter between your two compartments

and the boundary's dependence on it says something about whether the reservoir is driving the escape — but it's the least intuitive for clinicians. If you have room for two panels, dose and exchange rate together would cover both audiences well.

Normalize where you can. EC_{50}/MIC on a secondary top axis and dose expressed as a multiple of the standard regimen would make the whole plane dimensionless, which strengthens the math framing without costing the clinical reading.



pdf

Thank you. this was very very helpful. A friend of mine created this model with other math collaborators. (my friend has expertise in both medicine and math.) I'm going to upload the draft of his manuscript. Please assume the role of a math/medicine researcher with decades of experience building mathematical models that bridge that gap between math and medicine. He anticipates that he will submit it to a "math journal." 1) If you were a reviewer for this manuscript, what would your comments be? Please prioritize them into red and yellow flags; also indicate attributes of the paper that are excellent and/or "spot on." 2) If you were an expert in this field, what would you expect to see in this manuscript that is not presently there? 3) Please also provide reframing suggestions to increase the likelihood that this manuscript would be appreciated by researchers in this field and/or would advance knowledge in this field.

Orchestrating comprehensive critique across mathematical rigor and clinical relevance

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This is a substantial review, and there's a lot here — including some things that need fixing before it goes anywhere. Let me check what document tooling is available and then put the report together properly.

Prepared comprehensive referee report across mathematical and medical domains

Prepared comprehensive referee report across mathematical and medical domains

No markdown skill here, so I'll write it directly. Let me put together the referee report.

Created a file, read a file

Created a file, read a file

The full report is above. A few things I want to flag directly, because they'll matter more than the rest combined:

There's an unrecognized theorem in the paper. Your friend's most interesting observation — that R_b and R_{res} always escape together, across all four analyses — isn't a coincidence he stumbled on

four times. Near the extinction state, the resistant subsystem linearizes to a 2×2 matrix with strictly positive off-diagonal entries (the transfer terms). That matrix is irreducible and quasi-positive, so Perron–Frobenius guarantees a single dominant eigenvalue with a strictly positive eigenvector — meaning the two compartments *cannot* have independent boundaries. Both grow or both decay. It's a structural necessity, it's provable in about a page, and it converts his empirical anecdote into the result that would make a math journal want the paper. The same eigenvalue also explains, in one line, why the transfer rates did nothing until 10^{-3} h^{-1} .

The comparative sensitivity claim doesn't hold, and the fix is more interesting than the claim. All three reported escape fractions (34.7%, 53.9%, 30%) are recoverable to within two points from a normal CDF using only the threshold, the median, and the chosen CV. The Monte Carlo isn't measuring model sensitivity — it's reporting where he placed each distribution. But the arithmetic reveals something better: every threshold sits within about half a standard deviation of its assumed median. The infection is poised at criticality in every dimension examined. That's a real and defensible finding.

His headline clinical claim contradicts his stated result. The paper warns that negative blood cultures give false reassurance while bone stays infected — which requires compartment *discordance*. But the Results assert perfect *concordance*. Figure 2's caption mentions reservoir-only escape bins, so discordance does occur. Working the quasi-steady state, occult persistence lands around 5 CFU/mL in blood, just under his 10 CFU/mL LOD. Making that false-reassurance region an explicit object of study would turn his best clinical line from an assumption into a derived result.

And the reconciliation problem is serious. Three incompatible growth-rate sets, a Discussion reporting E_{max} thresholds five-fold different from the Results, equation (6) missing its transfer terms so it can't generate Figure 3A, and [cite: 1] placeholders scattered through the Discussion. A referee won't be able to verify a single number. That, unfortunately, will drive the decision no matter how good the underlying work is — so it's worth a dedicated pass before anything else.

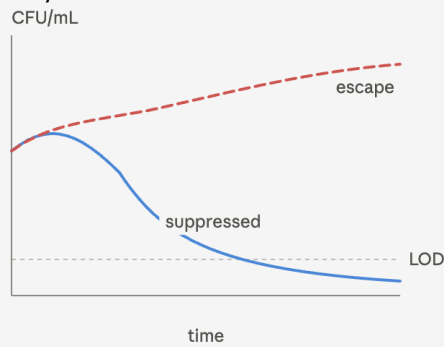
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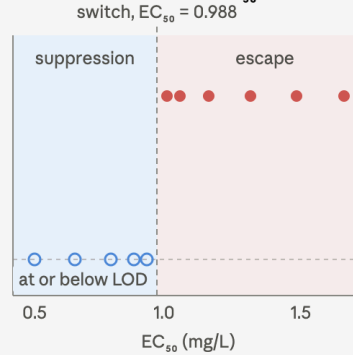
please provide the report in a format i can open in msword or google docs.

Here's the layout I'd argue for instead:

A dynamics either side of the switch



B terminal burden vs EC_{50}



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Schematic two-dimensional phase diagram of EC_{50} against dose

